

Software Analysis and Requirement Specification (SRS) Document

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1. Introduction

1.1. Purpose of Document

This Software Requirements Specification (SRS) document is intended to provide a comprehensive overview of the requirements for the development of a **School Management Platform (ERP)** called **[Platform_Name]**. The document covers the functional and non-functional requirements of the software, detailing the specifications for each module and submodule within the system.

Product Identification:

- **Product Name:** [Platform_Name]: School Management Platform (ERP)
- **Version:** 1.1.0 (Second Release)

1.2. Document Conventions

This document follows specific standards and typographical conventions to maintain consistency and clarity throughout the document. Below are the conventions used:

i. Formatting and Style:

- **Font Family:** The primary font used in this document is **Time New Romans**. For specific headings or design elements, the **Bricolage Grotesque** font may be used.
- **Font Sizes:**
 - **Body Text:** Standard body text is set to 12pt.
 - **Headings:** Main headings are set to 16pt and Subheadings are sized proportionately smaller and are bolded for emphasis.
 - **Emphasis:** Important terms and definitions are **bolded**.
- **Spacing:** Sections and elements are spaced 1.5 consistently to ensure readability and a clean layout.

ii. Requirements Notation:

- **Shall:** Indicates a mandatory requirement.
- **Should:** Indicates a recommended but not mandatory requirement.
- **May:** Indicates an optional requirement.
- **Will:** Describes future actions or events related to the system.

iii. Requirements Identification:

- Each requirement is uniquely identified with a prefix related to its module (e.g., **USR-001** for User Module, **FIN-001** for Financial Management Module, **STD-001** for student module, etc.).
- Requirements are categorized based on priority:
 - **High Priority (H):** Must be implemented for the system to function.
 - **Medium Priority (M):** Important but not essential for the initial release.
 - **Low Priority (L):** Nice-to-have features that may be included in future releases.

iv. Inheritance of Priorities:

- Higher-level requirements assume the priority of their detailed sub-requirements unless otherwise specified.

v. References and Cross-Referencing:

- References to other sections within the document are indicated by section numbers (e.g., 1. and sub section with decimal notation like 1.1).
- External references or documents are listed in the References section at the end of the document.

vi. Glossary:

- Technical terms and acronyms used within the document are defined in the Glossary section to ensure understanding across all stakeholders.

vii. Use of Graphics and Tables:

- Graphics and tables are used where necessary to provide visual clarity and are labeled appropriately for reference.

1.3. Intended Audience and Reading Suggestions

This SRS document is designed to be a comprehensive guide for all stakeholders involved in the development, implementation, and use of the [Platform_Name] School Management Platform (ERP). The document provides detailed information tailored to the needs of various audiences, including developers, project managers, marketing staff, end-users, QA teams, and documentation writers.

1.3.1. Intended Audience and description

ID	Stakeholder	Description
S-1	Documentation Writers	Documentation writers will use this SRS to create user manuals, help guides, and training materials. The document provides the necessary technical details and feature descriptions to create comprehensive documentation
S-2	Development team	Developers will use this document to understand the technical and functional requirements necessary to build the system (External Interface, functional and non-functional requirements). Detailed module specifications, data flow diagrams, and technical constraints are provided to guide development work
S-3	QA team	Quality Assurance teams will find the requirements and use cases particularly useful for designing test cases and ensuring that the system meets the specified criteria.
S-4	Project Managers	Project managers will reference this document to oversee the project's progress and ensure that all requirements are being met. This includes

		tracking milestones, managing scope, and ensuring that the project aligns with stakeholder expectations.
S-5	User	Although the document is primarily technical, certain sections, such as user interface designs and feature lists, are useful for end-users to understand how the system will address their needs.
S-6	Marketing Staff	Marketing teams can use this document to gain insights into the system's features and benefits, which will help in developing marketing strategies and communication plans that resonate with the target audience.

1.3.2. Reading Suggestions:

- **Overview Sections (Recommended for All Stakeholders):**

- **Introduction:** Start with the Introduction to understand the system's purpose, scope, and key objectives.
- **System Overview:** Provides a high-level view of the system architecture, modules, and key features.

- **Sections for Specific Stakeholders:**

- **Developers:**

- Focus on the **Functional Requirements** section ([Section 5](#)) for detailed system features and the **External Interface Requirements** ([Section 6](#)) to understand system interactions.
- The **Non-Functional Requirements** ([Section 7](#)) will also be critical for ensuring performance, security, and other quality attributes.

- **Project Managers:**

- Begin with the **Overall Description** ([Section 2](#)) to understand the system's context and user environment and **Project Strategy** ([Section 3](#)), which cover project objectives, constraints, and milestones.
- They should also review the **Assumptions and Dependencies** (Section 2.7) to understand potential risks and external factors affecting the project.
- **Marketing Staff:**
 - Focus on the **Product Features** (Section 2.2) and **User Classes and Characteristics** (Section 2.3) to gain insights into the system's capabilities and target audience, aiding in developing marketing strategies.
- **End-Users (Teachers, Students, Admin Staff, Parents):**
 - Find the **User Interfaces** (Section 6.1) and **System Features** (Section 5) sections most relevant, as they outline how the system will function from a user perspective and detail specific features designed to meet their needs.
- **QA Teams:**
 - Closely examine the **Functional Requirements** ([Section 5](#)) and **Non-Functional Requirements** ([Section 7](#)) to develop comprehensive test cases. The **Acceptance Criteria** for each feature should be detailed here for validation purposes.
- **Documentation Writers:**
 - use the **User Documentation** (Section 2.6) and **Glossary** (Appendix A) sections to gather the information necessary to create user manuals, help guides, and other support materials.

1.3.3. Document Organization:

The SRS document is structured to provide a logical flow from high-level overviews to detailed specifications:

1. Introduction
2. Overall Description (Project definition)
3. Project Strategy
4. Tools and Technology to Use
5. Functional Requirements and System features
6. External Interface Requirements
7. Non-Functional Requirements
8. Other Requirements
9. Appendices

1.3.4. Suggested Reading Sequence:

1. Introduction (Section 1)
2. Overall Description (Section 2)
3. Project Strategy (Section 3)
4. Functional Requirements and System Features (Section 5)
5. External Interface Requirements (Section 6)
6. Non-Functional Requirements (Section 7)
7. Other Requirements (Section 8)
8. Appendices (Appendix A-C)

1.4. Product Scope

The [Platform_Name] School Management Platform (ERP) is a comprehensive software solution designed to streamline and enhance the administrative, user management, financial management, human resources, course and performance management, services and products management, extracurricular activities, settings, maintenance and extracurricular functions within an educational institution. The platform aims to centralize various school management activities into a cohesive system, providing a user-friendly interface for students, parents, teachers, staff, and administrators.

- User management And School Management:
 - The System shall allow school's creation and Management (primary, secondary and high education)
 - The system shall allow RBAC (Role base access control) within each school (Dynamical role creation)
 - The system will allow parents to view their child's academic records (personal informations of users, enrollment information, academic performance course details, attendance records, extracurricular records, transcript, assessment and evaluation, financial and graduation information) , attendance, overview of school fees information and all financials informations, school events and communicate with teachers
 - The system will allow teachers to manage class rosters and student information
 - The system will allow teachers create and grade assignments and assessments
 - The system will allow teachers track student attendance and behavior
 - The system will allow teachers communicate with students, parents and other staff
 - The system will allow teachers access and update curriculum materials
 - The system will allow teachers create and update various events
 - The system will allow teachers generate and submit various reports
 - The system will allow Student Enrollment
 - The system will allow Personal Information Management for students
 - The system will allow Academic Records Management for students
 - The system will allow Student's attendance Tracking
 - The system will allow Student's performance Evaluation
 - The system will allow Reporting and Analytics
- Financial modules:
 - Tracking and managing income from various sources, including tuition fees and donations.

- Recording and managing school expenses, such as salaries, utilities, and maintenance.
- Budget creation, management, and comparison with actual expenses.
- Product and service management, including inventory tracking and sales.
- Generating daily and overall weekly financial reports for analysis and decision-making by authorized users, allowing them to filter the reports based on days(default), weeks, months, or customized periods.

This document will serve as a reference for developers, testers, and stakeholders to ensure that the software meets the specified requirements and functions as intended. The focus is on delivering a system that supports the daily operations of educational institutions, streamlines administrative tasks, and enhances communication between students, parents, teachers, and administrative staff.

1.4.1. Purpose and Objectives:

- **Schools Abstraction:** Each school should have his private space that he can manage his staff and perform his operations with an ease
- **Streamlining Administrative Processes:** The platform will automate routine administrative tasks such as student enrollment, attendance tracking, and timetable scheduling, reducing the workload on school staff and increasing operational efficiency.
- **Enhancing User's Communication and Collaboration:** By integrating chat functionality and other communication tools, the platform will facilitate real-time communication between teachers, students, parents, and administrators, fostering a collaborative educational environment. restriction: students and parents can't communicate directly with the admin.
- **Improving Financial Management:** The platform will offer robust financial management tools, including income and expense tracking, budget creation, and financial analysis, to ensure transparent and effective management of the school's finances.

- **Supporting Academic Excellence:** Through features like course and performance management, examination scheduling, and discipline tracking, the platform will help maintain high academic standards and monitor student progress effectively.
- **Managing Human Resources Efficiently:** The HR module will simplify tasks related to staff management, such as leave management, payroll processing, and time table scheduling, thereby improving HR operations within the institution.
- **Facilitating Extracurricular Activities:** The platform will also manage and track extracurricular activities, including school events, clubs, and sports, encouraging holistic student development.
- **Increase Productivity:** school administrators can spend less time maintaining records and sorting through unorganized data.
- **Save Natural Resources:** By keeping track of data digitally, your school can protect natural resources and avoid creating messy records that have to be reorganized at a later stage.
- **Increased Transparency with Parents:** Parents can check on their kids from time to time and keep track of their academic progress. This leads to improved transparency between parents and their children.

1.4.2. Relevance to Corporate Goals:

[Platform_Name] aligns with the broader corporate strategy of delivering innovative, technology-driven solutions that enhance the efficiency and effectiveness of educational institutions. By adopting this platform, educational institutions can ensure that they are leveraging modern tools to provide a high-quality educational experience, attract and retain talented staff, and engage effectively with students and parents.

1.4.3. Vision and Scope:

This SRS document outlines the Second iteration of the [Platform_Name] School Management Platform, with a focus on developing core modules such as **Schools Managements, User Management**, Financial Management, and Course and Performance Management. Future iterations will expand on these foundations, adding more features and enhancing system capabilities in line with user feedback and evolving institutional needs. A separate Vision and Scope document will provide a detailed roadmap for these future developments.

1.5. References

- <https://www.teachngo.com/blog/9-must-haves-for-School-ERP>
- *IEEE Recommended Practice for Software Requirements Specifications IEEE Computer Society, 1998*

2. Overall Description

2.1. Product Perspective (objective)

[Platform_Name] is a new, self-contained software product designed to address the growing need for a comprehensive, integrated system that can manage various administrative, financial, academic, and extracurricular activities within educational institutions. This platform is not a follow-on member of a product family, but rather a fresh, innovative solution that aims to replace disparate and often disconnected systems currently in use by schools.

Context and Origin:

- **Standalone Product:** The School Management Platform is designed as a standalone system, meaning it is not dependent on any existing platforms or systems. It is developed to be a comprehensive solution that can be implemented independently within an educational institution.
- **Replacement for Existing Systems:** Many schools currently use multiple software applications to manage different aspects of their operations, such as separate tools for

attendance, financial management, and student performance tracking. This platform is intended to consolidate these functionalities into a single, cohesive system, thereby eliminating the need for multiple software products and reducing complexity for users.

- **Component of a Larger System:** While the platform is a complete solution on its own, it is also designed to be compatible with other educational tools and systems. It can integrate with existing Learning Management Systems (LMS), content delivery networks, and other third-party applications through standard APIs and connectors.

System Interfaces:

- **User Interfaces:** The platform includes various user interfaces tailored to different school types, user roles, such as students, parents, teachers, staff, administrators, and user's group defined by the school. Each interface is designed to provide a user-friendly experience, with intuitive navigation and access to relevant features.
- **External Interfaces:** The platform is designed to interact with external systems through secure APIs. These interfaces allow the platform to exchange data with external tools like payment gateways, attendance tracking devices, and learning management systems.

Subsystem Interconnections:

The platform is modular, with different subsystems for user management, financial management, course and performance management, and other key functions. These subsystems are interconnected, allowing data to flow seamlessly across different parts of the platform. For example, student performance data from the course management subsystem can be accessed by the financial management subsystem to generate fee invoices based on academic performance.

Mermaid code architecture: Take it to <https://mermaid.live/edit> to have a look

```
graph TD
  %% Root
  Root["<b>[PLATFORM_NAME] ERP</b><br>Multi-Tenant SaaS Foundation"]

  %% Level 1 Modules
  M1["1.0 Core Admin & Identity"]
  M2["2.0 Academic Engine"]
  M3["3.0 Financial Management"]
```

M4["4.0 HR & Staff Management"]
M5["5.0 Communication & Portals"]

Root --> M1
Root --> M2
Root --> M3
Root --> M4
Root --> M5

%% Submodules Level 2 & 3: Module 1

M1 --> M1_1["1.1 Tenant & School Mgmt"]
M1_1 --> M1_1_1["1.1.1 Institution Tier Setup"]
M1_1 --> M1_1_2["1.1.2 Subdomain & White-Label"]
M1_1 --> M1_1_3["1.1.3 Campus & Calendars"]

M1 --> M1_2["1.2 Dynamic RBAC"]
M1_2 --> M1_2_1["1.2.1 Permission Catalog"]
M1_2 --> M1_2_2["1.2.2 Custom Role Builder"]
M1_2 --> M1_2_3["1.2.3 Membership Binder"]

M1 --> M1_3["1.3 User Directory"]
M1_3 --> M1_3_1["1.3.1 Student Registry"]
M1_3 --> M1_3_2["1.3.2 Parent-Student Links"]
M1_3 --> M1_3_3["1.3.3 Staff Profiles"]

%% Submodules Level 2 & 3: Module 2

M2 --> M2_1["2.1 Curriculum Operations"]
M2_1 --> M2_1_1["2.1.1 Subject Catalog"]
M2_1 --> M2_1_2["2.1.2 Cohort & Class Allocation"]
M2_1 --> M2_1_3["2.1.3 Course-Teacher Links"]

M2 --> M2_2["2.2 Assessment & Grading"]
M2_2 --> M2_2_1["2.2.1 Multi-Tier Grading Scales"]
M2_2 --> M2_2_2["2.2.2 Assessment Engine"]
M2_2 --> M2_2_3["2.2.3 Report Cards & Transcripts"]

M2 --> M2_3["2.3 Attendance & Behavior"]
M2_3 --> M2_3_1["2.3.1 Attendance Register"]
M2_3 --> M2_3_2["2.3.2 Disciplinary Records"]
M2_3 --> M2_3_3["2.3.3 Absence Alerts"]

%% Submodules Level 2 & 3: Module 3

M3 --> M3_1["3.1 Billing & Receivables"]
M3_1 --> M3_1_1["3.1.1 Fee Structures"]
M3_1 --> M3_1_2["3.1.2 Automated Invoicing"]
M3_1 --> M3_1_3["3.1.3 Payment Gateway & Cashier"]

M3 --> M3_2["3.2 Budgeting & Payables"]
M3_2 --> M3_2_1["3.2.1 Department Budgets"]
M3_2 --> M3_2_2["3.2.2 Operational Expenses"]

M3 --> M3_3["3.3 Products & Inventory"]
M3_3 --> M3_3_1["3.3.1 Uniform/Book Stock"]
M3_3 --> M3_3_2["3.3.2 Sales & POS"]

M3 --> M3_4["3.4 Reporting & Ledgers"]
M3_4 --> M3_4_1["3.4.1 Periodic Revenue"]
M3_4 --> M3_4_2["3.4.2 Outstanding Balances"]
M3_4 --> M3_4_3["3.4.3 Cash-Flow Statements"]

%% Submodules Level 2 & 3: Module 4

M4 --> M4_1["4.1 Staff Records"]
M4_1 --> M4_1_1["4.1.1 Dossiers & Credentials"]
M4_1 --> M4_1_2["4.1.2 Document Repository"]

M4 --> M4_2["4.2 Leave & Substitutes"]
M4_2 --> M4_2_1["4.2.1 Leave Approvals"]
M4_2 --> M4_2_2["4.2.2 Substitute Routing"]

M4 --> M4_3["4.3 Scheduling"]
M4_3 --> M4_3_1["4.3.1 Timetable Generator"]
M4_3 --> M4_3_2["4.3.2 Teacher Availability Matrix"]

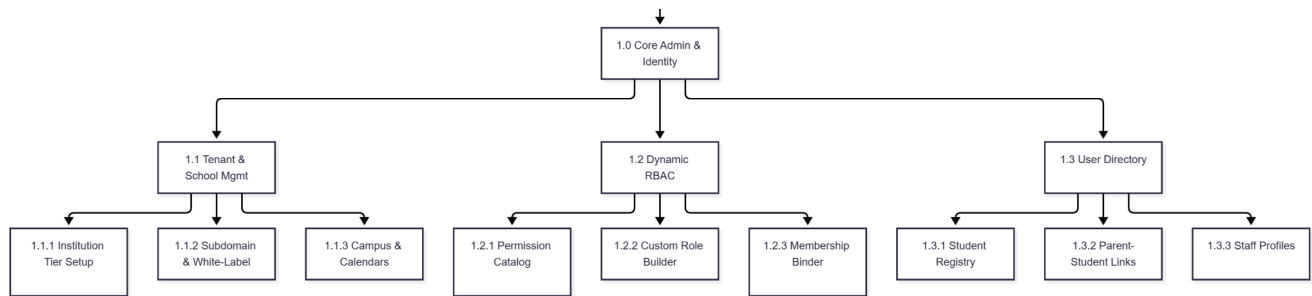
M4 --> M4_4["4.4 Payroll"]
M4_4 --> M4_4_1["4.4.1 Salary & Deductions"]
M4_4 --> M4_4_2["4.4.2 Payslip Dispatcher"]

%% Submodules Level 2 & 3: Module 5

M5 --> M5_1["5.1 Communication Hub"]
M5_1 --> M5_1_1["5.1.1 Real-Time Chat"]
M5_1 --> M5_1_2["5.1.2 Admin Direct-Chat Firewall"]
M5_1 --> M5_1_3["5.1.3 Multi-Channel Broadcasts"]

M5 --> M5_2["5.2 Stakeholder Portals"]
M5_2 --> M5_2_1["5.2.1 Parent Portal"]
M5_2 --> M5_2_2["5.2.2 Student Workspace"]
M5_2 --> M5_2_3["5.2.3 Teacher Workspace"]

M5 --> M5_3["5.3 Extracurricular"]
M5_3 --> M5_3_1["5.3.1 Clubs & Sports Rosters"]
M5_3 --> M5_3_2["5.3.2 Master Event Calendar"]



Root: Multi-Tenant SaaS Foundation

At the very top sits the shared platform layer. It powers white-labeling, isolated tenant environments, data integrity (such as PostgreSQL Row-Level Security), and platform-wide configuration. Every school operating on the platform executes within this single foundation while maintaining strict operational isolation.

1.0 Core Administration & Identity Management

This branch manages how tenants, users, and authorizations are initialized and governed across the ERP.

- **1.1 Tenant & School Management:** Configures each school's distinct institutional profile. It handles structural differences across tiers (Primary, Secondary, Higher Education), enables custom domains/subdomains (e.g., school.platform.com), and defines multi-campus terms and calendars.
- **1.2 Dynamic RBAC & Security:** To prevent breaking core portal workflows while allowing schools complete customization, the system uses a **two-layer RBAC structure**:
 - Layer A: User Archetype / Base System Role: This defines the structural identity of the user within the school and cannot be deleted or radically mutated because the frontend navigation depends on it:
 - **Student:** Read-only access to enrolled courses, timetables, and submissions.
 - **Parent:** Read-only access linked strictly to their associated children's data.
 - **Staff / Teacher:** Access to rosters, grading tools, and classroom registers.

- **Admin:** Full school-level administrative privileges.
- Layer B: Custom Dynamic Roles (Privilege Extension) : School administrators can create custom roles by bundling granular permissions. These roles are assigned to staff members to extend or specialize their administrative access without altering their base profile:
 - *Head of Department (HOD):* Can view and approve grades across an entire subject department.
 - *Bursar / Cashier:* Can access billing, cashiering, and financial reports, but not academic records.
 - *Discipline Officer:* Can log expulsions, suspensions, and incident logs school-wide.
 - *Exam Officer:* Can lock/unlock grading periods and publish official report cards.
- **1.3 User Directory & Profile Lifecycle:** Maintains the system-of-record for people within an institution. It manages student matriculation profiles, complex many-to-many guardian relationships (handling multi-child families and shared guardians), and faculty/staff records.

2.0 Academic Engine

The core pedagogical operational hub, driving daily teaching, course management, and student progression.

- **2.1 Curriculum & Course Operations:** Structures syllabus catalogs, credits, prerequisite rules, and cohort groupings. It binds subjects, classrooms, cohorts, and assigned teachers into active schedules.
- **2.2 Assessment & Performance Management:** Provides a configurable grading engine capable of processing GPA, letter grades, or percentage scales. It powers continuous assessment entry (quizzes, midterms, projects, exams) and aggregates them into official report cards and printable transcripts.

- **2.3 Attendance & Behavioral Tracking:** Captures real-time daily or period-level classroom attendance. It processes formal absence excuses, logs behavioral incidents/commendations, and triggers immediate alerts for unexcused absences.

3.0 Financial Management

Designed to handle both institutional receivables (tuition collection) and operational expenses with complete auditability.

- **3.1 Billing & Receivables:** Manages tuition schedules, localized fee levies, scholarships, and automated invoice dispatches. Integrates multiple payment rails (cashier counters, bank transfers, credit cards, mobile wallets) with instant receipt generation.
- **3.2 Budgeting & Payables:** Tracks departmental expenditures against approved institutional budgets, preventing operational overspending across utilities, facilities, and maintenance.
- **3.3 Products, POS & Inventory:** Functions as an internal retail register and inventory tracker for school stores-managing physical inventory for uniforms, textbooks, cafeteria operations, and supplies.
- **3.4 Financial Reporting & Audit:** Provides real-time financial roll-ups, daily cashier reconciliation sheets, aging reports for outstanding dues, and cash-flow summaries for administrative decision-making.

4.0 Human Resources & Staff Management

Handles the operational, contractual, and scheduling requirements of academic and non-academic staff.

- **4.1 Staff Records & Contract Management:** Serves as a digital HR dossier for teaching credentials, employment contracts, and compliance/accreditation documents.
- **4.2 Leave & Substitute Operations:** Orchestrates leave requests, accrual pools, and multi-tier manager approvals. When teachers take leave, it dynamically identifies schedule clashes and routes substitute teachers to affected classes.

- **4.3 Scheduling & Timetabling:** Resolves room allocation, teaching constraints, and teacher availability to generate conflict-free timetables for classes and faculty.
- **4.4 Payroll & Compensation:** Computes base pay, allowances, statutory deductions, and tax withholdings to generate recurring employee pay vouchers and disbursement logs.

5.0 Communication, Portals & Engagement

The stakeholder interaction layer, focused on structured collaboration while strictly maintaining administrative security boundaries.

- **5.1 Policy-Gated Communication Hub:** Enforces communication policies in real time. While it facilitates collaboration between teachers, parents, and students, it enforces an architectural firewall that **blocks students and parents from initiating direct messages with system administrators**. It also handles high-priority broadcast alerts (SMS, Push, Email).
- **5.2 Stakeholder Portals:** Provides tailored, role-specific web and mobile workspaces:
 - *Parent Portal:* A single dashboard for parents to track multiple enrolled children, review attendance, pay invoices, and contact teachers.
 - *Student Portal:* Central hub for study materials, timetables, assignment submissions, and published marks.
 - *Teacher Workspace:* Daily control center for logging attendance, submitting marks, and managing rosters.
- **5.3 Extracurricular & Institutional Events:** Manages athletics, student clubs, institutional event calendars, and non-academic achievements to ensure a comprehensive student record.

2.2. Product Features

The School Management Platform (ERP) incorporates a wide range of features designed to address the various needs of educational institutions. The following is a high-level summary of the major features of the platform:

Super Admin Architecture & Responsibilities

i. Tenant Provisioning & Lifecycle Management

- **School Onboarding & Setup:** Register new schools, allocate custom subdomains (e.g., `greensprings.platform.com`), and select the institutional tier (Primary, Secondary, Higher Education).
- **Tenant State Controls:** Temporarily suspend, archive, reactivate, or delete entire school accounts (e.g., for non-payment or end-of-contract).
- **Feature Flags & Module Activation:** Toggle specific modules on or off per school (e.g., enable the Cafeteria POS for School A, disable it for School B).

ii. SaaS Subscription & Platform Billing

- **Subscription Tier Management:** Define recurring pricing plans (e.g., Per-Student/Month, Tiered Annual Licensing, or Flat Enterprise).
- **Payment Collection & Invoicing:** Bill the schools directly for using the platform, track payment statuses, and trigger automated license suspensions for overdue accounts.
- **Usage Limits & Quotas:** Enforce storage caps (document uploads, homework files), bandwidth usage, and active student/staff count limits.

iii. Global Communication Rails & Integrations

- **Gateway Providers:** Configure platform-wide credentials for the WhatsApp Business API, SMS aggregators (e.g., Twilio, Infobip), email providers (SendGrid, Postmark), and mobile money gateways.
- **Credit Balance Monitoring:** Monitor messaging credits, per-tenant SMS/WhatsApp consumption, and set rate limits to prevent spam or API abuse.

iv. Impersonation & Tenant Support

- **"Login-as-School-Admin" (Impersonation Mode):** Securely log into any tenant's workspace with a single click to troubleshoot issues reported by school administrators—without needing to know their passwords.

- **Platform Helpdesk & Ticketing:** Manage support requests submitted by school admins regarding software bugs or configuration issues.

v. System Health, Global Auditing & Security

- **Global Audit Trails:** Track every critical platform event (e.g., tenant creation, credential changes, data purges, and Super Admin impersonation sessions).
- **Platform Analytics:** Real-time metrics on total active schools, aggregated student population, system load, uptime, and database storage utilization.
- **Data Backups & Compliance:** Trigger global and per-tenant database snapshots, export entire tenant data packages, and execute GDPR-compliant tenant purges.

1. Identity, Access & User Management

- **Authentication & Profile Management:** Multi-factor authentication, self-service password recovery, profile customization, and secure credential handling.
- **Two-Layer Dynamic RBAC:**
 - *Base Archetypes:* Fixed interface templates for **Admin**, **Staff/Teacher**, **Parent**, and **Student**.
 - *Custom Dynamic Roles:* Administrative creation of specialized institutional roles (e.g., Bursar, Head of Department, Discipline Master) with granular capability assignment.
- **Tiered Student Account Provisioning:**
 - *Primary / Elementary:* Record-only tracking; no login credentials issued.
 - *Secondary & Higher Education:* Direct student login for schedules, assignments, syllabus progression, and grades.
- **Parent & Multi-Child Linking:**
 - Multi-child session switcher allowing parents to toggle between siblings within the same institution.
 - Consolidated parent dashboards displaying attendance, academic results, discipline logs, and aggregated fee balances.

- **Staff Registry & Dossiers:** Academic and non-academic employee profiles, qualification archives, assigned responsibilities, and emergency contacts.

2. Academic & Course Performance Engine

- **Curriculum & Program Progression Tracker:**
 - Syllabus milestone tracking allowing teachers to log completed chapters, upload resources/files, and compare progress against term schedules.
 - Program completion analytics exportable as PDF/Excel for administrative oversight.
- **Classroom & Timetable Operations:**
 - Cohort management, section enrollments, student rosters, and room allocation.
 - Drag-and-drop or conflict-checked timetable scheduling for classes, teachers, and practical labs.
- **Assessments & Gradebook:**
 - Homework builder supporting file attachments, textbook page prompts, and online submissions.
 - Multi-tier grading scales (GPA, percentage, or letter grades) supporting quizzes, continuous assessments, and term finals.
 - Administrative grade override workflow with change-log tracking for accidental grade submissions.
 - Automated report card and official transcript generation with export options in PDF and Excel.
- **Attendance & Discipline:**
 - Period-by-period or daily attendance registers with pattern analysis (identifying chronic absentees).
 - Disciplinary tracking logging positive commendations, infractions, and formal sanctions.

3. Financial Management & Accounting

- **Tuition & Receivables:**

- Configurable fee structures (tuition installments, medical fees, lab fees, transportation, and feeding).
- Automated recurring invoice generation and multi-child sibling discounts.
- Payment cashiering and digital receipting across bank transfers, cash, credit/debit cards, and mobile money.
- **Expenditure & Payables:**
 - Operational expense tracking across utilities, maintenance, office supplies, and external vendor bills.
 - Staff payroll integration for base salary disbursements, statutory deductions, bonuses, and downloadable employee payslips.
- **Budgeting & Fiscal Reporting:**
 - Departmental budget allocations with threshold alerts against actual expenditures.
 - Real-time financial dashboards reporting daily cashier collections, fee-aging ledgers, and exportable weekly/monthly balance sheets.

4. Inventory, Auxiliary Services & POS

- **Physical Inventory & Sales:** Stock management, reorder thresholds, and point-of-sale logging for uniforms, textbooks, and stationery.
- **Service Management:**
 - *Transportation:* Route management, bus allocations, and recurring student transit billing.
 - *Cafeteria / Feeding:* Meal plan subscriptions, daily canteen service logs, and usage tracking.

5. Human Resources & Staff Operations

- **Leave Management:** Custom leave policies, digital leave request workflows, multi-step approvals, and substitute teacher routing to prevent unstaffed classes.
- **Staff Attendance & Performance:** Biometric or digital check-in tracking and teacher evaluation metrics.

6. Communication, Notifications & Portals

- **Priority-Based Dispatcher:**
 - **Primary Channel:** In-App/site, Automated WhatsApp messaging for instant fee reminders, emergency alerts, absence notices, and event updates.
 - **Fallback Channel:** Automatic fallback to standard SMS and email if WhatsApp delivery fails or is unreachable.
- **Policy-Gated Chat Hub:**
 - Real-time direct messaging between Teachers ↔ Parents and Teachers ↔ Students.
 - **Communication Firewall:** Hard application restriction preventing students and parents from initiating direct private chats with administrators.
- **Institutional Master Calendar:** Extracurricular scheduling, sports fixtures, club activities, and parent-teacher conference coordination.

2.3. User Classes and Characteristics (target audience)

User Class	Freq. of Use	Functions Used	Technical Expertise	Characteristics & Responsibilities
Super Admin (Platform Owner)	Daily (H)	Tenant provisioning, SaaS subscription billing, gateway configuration (WhatsApp/SMS), global audit logs, tenant impersonation.	Expert; systems & infrastructure operations.	Manages the SaaS platform above school tenants. Enforces quotas and oversees system availability without accessing tenant classroom records.

School Administrator	Daily (H)	Tenant configuration, dynamic RBAC/role creation, staff onboarding, academic calendar setup, report card locks, and high-level financial/academic overrides.	Advanced; administrative software experience.	Oversees operational governance within a single school. Possesses broad access rights, subject to immutable system audit trails.
Bursar / Financial Officer	Daily (H)	Fee structure configuration, budget planning, expenditure approvals, bank reconciliation, and fiscal balance sheet reporting.	Intermediate to Advanced; accounting domain knowledge.	Manages institutional fiscal compliance, reviews transactions posted by cashiering staff, and analyzes operating cash flow.
Accountant / Cashier	Daily (H)	Tuition collection, digital receipting, POS desk operation (uniforms/cafeteria), and expense voucher recording.	Basic to Intermediate; transactional data entry.	Routine financial recordkeeping. Cannot permanently delete posted ledger entries; corrections must follow void/reversal workflows.
Teachers & Faculty	Daily (H)	Gradebook management, daily attendance logging, syllabus milestone tracking, homework assignment, and parent messaging.	Intermediate; web and mobile application fluency.	Primary academic operators. Requires streamlined interfaces to minimize classroom friction and update curriculum timelines.

Support Staff (HR, Canteen, Transit)	Regular (M to H)	Inventory management, bus route rosters, canteen meal validation, leave applications, and shift timetables.	Basic to Intermediate; role-specific UI tasks.	Operates solely within delegated module views (e.g., transit drivers viewing route manifests, inventory clerks logging uniform sales).
Parents / Guardians	Weekly to Daily (M to H)	Multi-child academic monitoring, fee statement review, digital tuition payments, and direct messaging with teachers.	Basic; mobile and smartphone user.	Access is strictly scoped to linked dependents. Prohibited from initiating direct messaging threads with School Administrators.
Students (Secondary & Higher Ed)	Daily (H)	Assignment submission, timetable/room checking, personal grades and transcript viewing, and curriculum resource downloads.	Basic to Intermediate; digital mobile/web literacy.	Direct account access is restricted to secondary and tertiary students. Primary pupils do not hold login credentials.
System Integrators / Developers	Periodic (L to M)	Webhook and REST API management, LMS/SIS third-party syncing, external biometric device connectors.	Advanced; API standards and system integration.	Focuses strictly on technical infrastructure, automated data pipelines, and third-party API interoperability.

Quality Assurance (QA) Personnel	Intermittent (M)	Automated regression testing, RBAC permission penetration tests, schema migration verification, and staging sign-offs.	Advanced; software QA and testing methodologies.	Validates functional specifications in staging and sandbox environments that mirror multi-tenant production configurations.
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2.4. Operating environment | System Platforms

The School Management Platform (ERP) is designed to operate in a variety of environments, supporting different hardware and software configurations to ensure accessibility and usability across a wide range of users. Below is an overview of the intended operating environment and system platforms:

2.4.1. Hardware Platform

- **Client Devices:**
 - **Desktops and Laptops:** The platform will be accessible through modern web browsers on both Windows and macOS operating systems.
 - **Tablets and Smartphones:** The platform will be optimized for mobile devices, including iOS and Android systems, ensuring a responsive design that adapts to different screen sizes.
 - **Offline functionalities:** by saving data locally on computer or mobile phone so that data are available offline, or when there's data issues.
- **Server Requirements:**

- [The platform will be hosted on cloud-based servers, utilizing scalable infrastructure to accommodate varying levels of traffic and data storage needs.]

2.4.2. Operating Systems

Windows, macOS, iOS, Android

2.4.3. Web Browsers

Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Mobile browsers

2.4.4. Software Components

- **Database:** The platform will use SQL database management system.
- **Back-End:** The back-end will be developed using **Nest.js** with **TypeScript** for handling server-side logic.
- **Front-End:** The front-end will be developed using **NextJS** with **Typescript**, ensuring a dynamic and responsive user interface.
- **APIs:** RESTful APIs will be used to handle communication between the front-end and back-end systems.
- **Security Frameworks:**
 - The platform will incorporate **OAuth 2.0** for secure authentication and authorization.
 - **SSL/TLS** encryption will be implemented for data transmission.
- **Third-Party Integrations:**
 - Integration with payment gateways for handling financial transactions for school fees and all the rest.
 - Integration with external communication tools (e.g., email and SMS services) for parents, etc.

- The system must support data import/export functionalities using standard formats like **CSV** or **XML** for interoperability with other systems, export documents in PDF, Excel format.

2.4.5. Limitations and Constraints

- **Corporate and Regulatory Policies:** The platform must comply with GDPR and other relevant data protection regulations to ensure user privacy and data security.
- **Hardware Limitations:** The platform should be optimized to run efficiently on devices with limited processing power and memory, especially older devices used by students and parents.
- **Interfaces to Other Applications:** The platform should seamlessly integrate with existing school management systems, accounting software, and other third-party tools used by the institution.
- **Specific Technologies and Tools:** The development team will be required to use version control systems (e.g., Git), continuous integration tools, and deployment automation tools (Vercel) to maintain consistency and quality throughout the development process, product management tools including Jira, confluence, clickup. Sharing tools such as google docs, google sheet; Figma for the user interface design; And architect for diagrams..
- **Security Considerations:** Data security is paramount, and the platform must implement robust security measures, including regular security audits, vulnerability testing, and data encryption.
- **Design Conventions:** The platform should adhere to established design conventions to maintain consistency in user experience, such as Material Design principles or similar UI/UX standards.

2.5. Design and Implementation Constraints

In the development of the [Platform_Name] School Management Platform (ERP), several constraints must be considered that will affect the design and implementation of the software. These constraints are as follows:

- **Technologies, Tools and Database:**

- **Version Control:** Git will be used for version control, with GitHub as the repository hosting service.
- **Continuous Integration/Continuous Deployment (CI/CD):** Tools such as Vercel will be employed to automate testing and deployment processes.
- **APIs:** RESTful APIs will be used for communication between the front-end and back-end systems.
- **Primary Database:** PostgreSQL, an SQL database, will be used for its scalability and flexibility.
- **Backup/Secondary Database:** MySQL or PostgreSQL may be used if relational database functionalities are required. TODO: Let's make research on NOTION

- **Development Standards:**

- The software development process will adhere to **Agile methodologies** using **SCRUM** framework to ensure iterative development and timely delivery of features.
- **Code Quality:** Adherence to coding standards such as Storybook, atomic design for JavaScript will be mandatory to maintain code quality and consistency.

- **User Access and Permissions:**

- The platform must implement strict user access controls, ensuring that only authorized users can access sensitive data.
- **Business Rules:** All business logic related to school operations, such as student attendance, grading systems, and fee management, must be accurately reflected in the platform's functionalities.

Standard Data Exchange Formats

- **Data Formats:**
 - **JSON and Form-data** will be the primary format for data exchange between client-side and server-side components.
 - **XML** may be used for integrations requiring standard data interchange formats, particularly in communication with legacy systems:
<https://google.github.io/styleguide/xmlstyle.html>

Design Conventions or Programming Standards

- **Programming Standards:** The codebase will follow industry best practices, including modular design patterns, DRY (Don't Repeat Yourself) principles, and SOLID principles for object-oriented design.
- **Maintenance:** Clear documentation and well-structured code are essential to ensure that the platform can be easily maintained and extended by future developers.
- **Localization:** The platform should support multiple languages to cater to different regions, with a primary focus on English, but with the ability to add other languages as needed.

Security Considerations

- **Data Encryption:** All sensitive data will be encrypted at rest and in transit using industry-standard encryption methods such as **AES-256**.
- **Authentication and Authorization:** The platform will implement **OAuth 2.0** for secure authentication and role-based access control to restrict access based on user roles.
- **Regular Security Audits:** Periodic security assessments and vulnerability testing will be conducted to identify and mitigate potential security risks.

2.6. User Documentation

2.6.1. User Manuals

- **Admin User Manual:** A detailed guide tailored for administrators, covering system setup, user management, role assignments, configuration settings, and troubleshooting procedures.
- **Teacher/Staff User Manual:** Instructions on how to use the platform's features relevant to teachers and staff, including class management, grade entry, attendance tracking, and communication tools.
- **Student and Parent User Manual:** A simplified guide designed for students and parents, explaining how to navigate the platform, access grades, view attendance records, and communicate with school staff.

2.6.2. Online Help

- **Context-Sensitive Help:** Integrated help accessible directly from the platform's interface, providing users with relevant information based on their current task or screen.
- **FAQ Section:** A dedicated section within the platform addressing common questions and issues that users may encounter.

2.6.3. Tutorials

- **Video Tutorials:** Step-by-step video guides demonstrating the use of key features of the platform, catering to different user roles (e.g., administrators, teachers, students).
- **Interactive Walkthroughs:** Guided tours within the platform that walk users through essential tasks, such as setting up classes, entering grades, or making fee payments.

2.7. Assumption and dependencies

This section outlines the assumptions made during the development of the [Platform_Name] School Management Platform and identifies key dependencies that could impact the project's

success. These assumptions and dependencies must be carefully considered to ensure that the project remains on track and meets the requirements stated in this SRS.

Assumptions

- **Availability of Third-Party Components:** It is assumed that all third-party components, such as libraries, frameworks, and APIs, will be available and compatible with the platform's technology stack throughout the development cycle.
- **Stable Operating Environment:** The platform is assumed to be deployed in a stable and well-maintained IT infrastructure, with regular updates and support for the operating systems and browsers used by the platform's users.
- **User Proficiency:** It is assumed that end-users, including administrators, teachers, students, and parents, will have basic digital literacy and be able to navigate web-based applications with minimal training.
- **Timely Feedback and Collaboration:** The project assumes active and timely participation from stakeholders, including regular feedback and collaboration during the development process to align the platform with user needs and expectations.
- **Data Availability and Integrity:** It is assumed that all necessary data for initial setup, such as student records, staff details, course information, and financial data, will be provided accurately and in a timely manner by the institution.
- **Adherence to Regulatory Requirements:** The development of the platform assumes that all legal and regulatory requirements relevant to data protection, privacy, and educational standards are clearly communicated and remain stable during the project lifecycle.

Dependencies

- **Third-Party Software and Services:** The platform's functionality may rely on third-party services such as payment gateways, email/SMS services, and external APIs for data integration. The project is dependent on the continued availability, reliability, and support of these services.

- **Timely Delivery of Components:** The development process depends on the timely delivery of external components, including those developed by other teams or vendors, such as custom modules or integrations with existing systems.
- **Updates to Operating Systems and Browsers:** The platform's performance and compatibility are dependent on the support for the latest versions of operating systems and web browsers. Any delays or issues with updates may affect the system's functionality.
- **Hardware and Infrastructure Upgrades:** The project is dependent on the school or institution's ability to maintain or upgrade their hardware and network infrastructure to support the platform's requirements, including server capacity, internet bandwidth, and security protocols.
- **Compliance with Security Protocols:** The platform's security relies on the timely implementation and maintenance of security protocols, including SSL certificates, firewalls, and regular security audits, as provided by the hosting environment or institution.
- **User Training and Adoption:** Successful implementation and use of the platform depend on the institution's commitment to training users and encouraging adoption. The project assumes that adequate training resources and sessions will be made available to all user classes.

3. Project Strategy

3.1. Define Project Work Strategy

Write about your project work strategy. Describe the strategy of design and development software and software development life cycle.

Agile Development Methodology: iterative progress and Sprint Cycle (3 weeks for each sprint)
--

Modular design for each module

Phased Implementation: each phase delivering a set of functional modules and initial phase focus on core modules (the director module)
Quality assurance and testing: unit testing, integration testing, user acceptance testing (UAT), continuous integration and deployment (CI/CD) with vercel
Documentation and knowledge transfer: design documents, user manual, technical specifications
Risk management: identify potential risk early and develop mitigation strategies

3.2. Milestones of the Project

Prepare and write about the different milestones of the project. Describe the date of the deliverable here.

No. of Milestones	Delivery Date
Project Kickoff and planning (Requirements)	
Requirement specification and approval (SRS)	
Design phase completion (DB, UI/UX, architecture, prototypes) of Module 1: Director	
Prototype delivery for module 1	
Implementation/development completion Module 1: director	
...	
UAT for phase (module) 1	
Module 2	
Module 3	
Integration of all modules and System Testing	
Final User Acceptance Testing	
Training delivery	

Deployment	
Support, optimization	
Project cloture	
Maintenance	

4. Tools and Technology to Use

Tools and Technologies in the Project	
Programming Language	JavaScript, TypeScript
Web Server	Node.js, Nest.js
Database	PostgreSQL
Server OS	Linux, Windows server
Integrations	REST APIs, OAuth 2.0, email sending
Frameworks	NextJS, Redux-toolkit, Tailwind css, Docker
Additional tools	Git, Excel, JIRA, VSCode, Confluence, Google docs

5. Functional requirements (System features)

1.0 Platform Administration & Multi-Tenancy (Super Admin)

- **FR-PL-01 (Tenant Provisioning):** The system shall allow Super Admins to provision new school tenants, assign subdomains (`{tenant}.platform.com`), and configure institution tiers (Primary, Secondary, Higher Education).

- **FR-PL-02 (Lifecycle & Subscriptions):** The system shall support SaaS subscription tier configurations, automated tenant billing cycles, quota enforcement (storage, active student caps), and instant tenant suspension/reactivation.
- **FR-PL-03 (Tenant Impersonation):** The system shall allow authorized Super Admins to securely impersonate a School Administrator session for troubleshooting without accessing or exposing user passwords.
- **FR-PL-04 (Global Dispatch Rails):** The system shall centrally manage integration keys and credit balances for messaging backends (WhatsApp Business API, SMS aggregators, SMTP relays, payment webhooks).

2.0 Core Administration, Dynamic RBAC & Identity

- **FR-ID-01 (Two-Layer Dynamic RBAC):** The system shall enforce four base user archetypes (**Admin**, **Staff/Teacher**, **Parent**, **Student**) while allowing School Admins to dynamically create custom roles (e.g., Bursar, HOD, Discipline Master) bound to granular system permissions.
- **FR-ID-02 (Tiered Account Provisioning):**
 - The system shall provision student records **without** login credentials for Primary/Elementary tiers.
 - The system shall automatically generate distinct, secure login credentials for students in Secondary and Higher Education tiers upon enrollment.
- **FR-ID-03 (Student & Matriculation Registry):** The system shall generate unique matriculation numbers based on configurable school templates and manage student dossiers (demographics, medical records, emergency contacts, documents).
- **FR-ID-04 (Parent-Student Linkage):** The system shall support many-to-many parent-student relationships and provide a session-based child switcher allowing parents to monitor multiple enrolled siblings under a single login.
- **FR-ID-05 (Staff Onboarding):** The system shall maintain academic and non-academic staff dossiers, employment contracts, teaching certifications, and department allocations.

3.0 Academic Engine & Performance Management

- **FR-AC-01 (Curriculum & Course Mapping):** The system shall allow administrators to configure subject catalogs, credit hours, weighting coefficients, prerequisites, and assign courses to cohorts, sections, and designated teachers.
- **FR-AC-02 (Syllabus & Milestone Progression):** The system shall allow teachers to log chapter milestones and syllabus coverage dates, generating downloadable progress comparison reports against academic term calendars.
- **FR-AC-03 (Timetable Engine):** The system shall support conflict-checked timetable scheduling for classes, teachers, and specialty classrooms/laboratories.
- **FR-AC-04 (Assessment & Homework Workflow):** The system shall enable teachers to publish assignments (text, file attachments, book page prompts) and record grades for quizzes, assignments, practicals, and midterms/finals.
- **FR-AC-05 (Grading Scales & Transcripts):** The system shall calculate composite scores using customizable grading scales (letter grades, percentages, 4.0/5.0 GPA scales) and generate automated, printable report cards and official transcripts with PDF export.
- **FR-AC-06 (Grade Override & Audit):** The system shall provide an administrative grade override workflow that requires a logged justification reason and logs immutable before/after values.
- **FR-AC-07 (Attendance Management):** The system shall allow teachers to take period-level or daily classroom attendance (Present, Absent, Late, Excused) and generate chronic absenteeism alerts.
- **FR-AC-08 (Discipline Tracking):** The system shall record student commendations, disciplinary infractions, sanctions, and suspensions, linking incidents directly to student profiles.

4.0 Financial Management & POS

- **FR-FN-01 (Fee Structuring & Invoicing):** The system shall automate the generation of term/installment invoices based on configurable fee schedules, recurring levies, scholarships, and sibling discounts.
- **FR-FN-02 (Multi-Rail Cashiering):** The system shall process payments via integrated online payment rails, bank transfers, mobile money, and physical cashier desks, instantly issuing tamper-proof digital receipts.

- **FR-FN-03 (Ledger Immutability):** The system shall prohibit the deletion of completed financial transactions; adjustments must strictly follow void/reversal journal entries backed by audit logs.
- **FR-FN-04 (Budgeting & Expense Payables):** The system shall track departmental operational expenses (utilities, supplies, maintenance) against pre-approved budgets and trigger overdraft alerts.
- **FR-FN-05 (Inventory & POS Sales):** The system shall track stock levels, reorder thresholds, and point-of-sale transactions for school uniforms, textbooks, cafeteria meal vouchers, and stationery.
- **FR-FN-06 (Financial Analytics):** The system shall provide real-time filterable financial reports (daily cashier collections, fee-aging reports, and weekly/monthly revenue vs. expenditure balance sheets).

5.0 Human Resources & Staff Operations

- **FR-HR-01 (Leave Request Workflow):** The system shall process leave requests, track entitlement balances, and route submissions through multi-tier administrative approval hierarchies.
- **FR-HR-02 (Substitute Teacher Routing):** The system shall automatically flag timetable conflicts when a teacher takes approved leave and recommend available substitute teachers for unstaffed periods.
- **FR-HR-03 (Payroll & Deductions):** The system shall compute monthly payroll based on salary scales, allowances, and statutory deductions, generating downloadable payslips for employees.
- **FR-HR-04 (Staff Attendance):** The system shall log daily staff check-in/check-out records and compile staff attendance metrics for evaluation.

6.0 Communication, Engagement & Portals

- **FR-CM-01 (Communication Firewall):** The system shall enforce an architectural boundary that strictly prohibits Parents and Students from initiating direct messaging threads with School Administrators.
- **FR-CM-02 (Permitted Messaging):** The system shall provide real-time messaging between Teachers and Parents, and Teachers and Students (for secondary/tertiary tiers).

- **FR-CM-03 (Priority Notification Dispatcher):** The system shall dispatch high-priority notifications (emergency notices, absence alerts, fee payment reminders) via **WhatsApp Business API as the primary channel**, automatically falling back to **SMS and Email** upon failure.
- **FR-CM-04 (Extracurricular & Master Calendar):** The system shall manage club rosters, sports team participation, and maintain an institutional master calendar for term dates, exams, and parent-teacher meetings.
- **FR-CM-05 (Role-Specific Portals):** The system shall provide dedicated dashboards:
 - *Parent Portal:* Consolidated multi-child view for invoices, payment receipts, attendance, and teacher communications.
 - *Student Portal:* Enrolled class schedules, syllabus trackers, assignment submission portals, and released grades.
 - *Teacher Portal:* Daily rosters, grade entry sheets, attendance registers, and syllabus management.

6. External Interface Requirements

6.1. User Interfaces

The user interface (UI) of the School Management Platform (ERP) will be designed to be intuitive, user-friendly, and consistent with modern design principles. It will include logical characteristics that cater to different user roles, including students, parents, teachers, staff, and administrators. The UI will adhere to a consistent design style, ensuring a smooth user experience across all modules.

❖ Screen Layout and Design

- **GUI Standards:** The platform will follow a flat design aesthetic with minimalistic elements, ensuring clarity and ease of use.
- **Screen Layout Constraints:** Screens will be designed to be responsive, allowing for accessibility across various devices (desktop, laptop, tablet, smartphone).
- **Standard Buttons and Functions:**

- **Navigation:** Common navigation buttons like "Home," "Back," "Next," "Save," "Cancel," and "Logout" will be standardized across all screens.
- **Help:** A dedicated "Help" button/icon will be available on all screens, providing context-sensitive assistance.
- **Search:** A global search bar will be available on the header of each page to quickly locate information.
- **Notifications:** A bell icon for notifications will be placed in the top-right corner of the screen, indicating new alerts or messages.
- **User Profile:** A user profile dropdown menu will be available for quick access to settings and account management.

❖ **Keyboard Shortcuts**

Common keyboard shortcuts will be implemented to enhance usability, such as:

- **Ctrl + S:** Save
- **Ctrl + F:** Search
- **Ctrl + P:** Print
- **Esc:** Cancel/Close

❖ **Error Message Display Standards**

Error messages will be displayed in a non-intrusive manner, with clear descriptions of the problem and possible solutions. They will include an "X" icon to close the message.

❖ **Software Components Needing a User Interface**

- **Dashboard Interface:** For displaying key metrics and notifications.
- **User Management Interface:** For managing user roles, profiles, and access rights.
- **Financial Management Interface:** For managing and tracking financial transactions, budgets, and reports.
- **Course and Performance Management Interface:** For managing courses, exams, student performance, and reports.
- **Communication Interface:** For chat functionality, notifications, and messaging.

6.2. Hardware Interfaces

The School Management Platform (ERP) will interact with various hardware components, requiring clear definitions of both logical and physical characteristics.

Supported Device Types

- **Servers:** The platform will be hosted on cloud-based servers with support for Linux and Windows operating systems.
- **Client Devices:** Users will access the platform via desktop computers, laptops, tablets, and smartphones running on popular operating systems.
- **Printers:** Integration with network printers for printing reports, receipts, and other documents.
- **Scanners:** Support for document scanners to digitize and upload student records and other documents.

Data and Control Interactions

- **Data Flow:** Data will be transmitted between client devices and the server using HTTPS for secure communication.
- **Control Interactions:** User actions / events (e.g., clicking a button, submitting a form) on client devices will trigger corresponding processes on the server.

Communication Protocols

- **HTTP/HTTPS:** For web-based communication between the client devices and the server.
- **USB/Network Connections:** For connecting to printers and scanners.

6.3. Software Interfaces

The platform will integrate with various software components, including databases, operating systems, and third-party tools.

Operating Systems

- **Server OS:** The platform will be compatible with both Linux (e.g., Ubuntu, CentOS) and Windows Server environments.
- **Client OS:** Compatibility with Windows, macOS, Linux, Android, and iOS.

Database: The platform will use MongoDB as the primary database.

Third-Party Tools and Libraries

- **Payment Gateways:** Integration with payment gateways like MoMo, OM, PayPal, for managing online transactions.
- **Email and SMS:** Integration with email servers (e.g., SMTP) and SMS gateways for notifications.
- **Authentication Services:** Integration with third-party authentication services (e.g., OAuth, LDAP) for user login and management.
- **Reporting Tools:** Integration with reporting tools like JasperReports or custom-built solutions for generating reports.

Dashboard:

- Overview of key metrics and alerts.
- Navigation panel for accessing different admin functionalities.

User Management:

- Add, edit, and delete user accounts.
- Search and filter options for managing user roles.

Data Management:

- Import/export data functionality.
- Data visualization tools for analysis.

Buttons and Functions:

- **Save/Update:** to save changes made.
- **Delete:** to remove items or records.
- **Search:** to find specific information.
- **Filter:** to refine data based on criteria.
- **Generate Report:** to create detailed reports.

Messages:

- **Success/Error Messages:** for user actions.
- **Confirmation Dialogs:** for critical operations.
- **Informational Messages:** for system updates or alerts.

Style Guides:

- **Color Scheme:** Use school colors for branding.
- **Typography:** Clear and readable fonts.
- **Consistent Layout:** Maintain uniformity across screens.

Responsive Design: Ensure usability on various devices.

6.4. Communications Interfaces

The platform requires robust communication interfaces to handle various interactions, including network protocols, email, and browser-based communications.

Email and Notifications

- **Protocols:** The platform will use SMTP for sending emails. IMAP/POP3 may be used for retrieving emails.
- **Format:** Emails will be sent in HTML and plain text formats, depending on the recipient's preferences.
- **Security:** Emails will be encrypted using SSL/TLS to ensure secure communication.

Web Browser Communication

- **Web Protocols:** HTTP/HTTPS will be used for all web-based communications, ensuring secure data transmission between clients and the server.
- **Browsers Supported:** Compatibility with major web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
- **WebSockets:** For real-time communication features like chat functionality, the platform will use WebSocket protocols.

Network Server Communication

- **Protocols:** TCP/IP will be used for internal network communications. The platform will also support FTP/SFTP for file transfers between servers and client devices.
- **Data Transfer Rates:** The system will be optimized for varying network conditions, with support for both high-speed and low-bandwidth connections.
- **Synchronization:** Mechanisms like AJAX and polling will be used to keep client interfaces updated with real-time data from the server.

Security and Encryption

- **SSL/TLS:** All communication will be secured using SSL/TLS encryption to protect data integrity and confidentiality.
- **Firewalls and Intrusion Detection:** The system will include support for firewalls and intrusion detection systems to safeguard against unauthorized access.

7. Non-Functional Requirements

7.1. Security

- The system shall encrypt all financial data at rest and in transit.
- The system shall implement multi-factor authentication for users with access to sensitive financial data.
- The system shall ensure role based access and user permission when accessing sensitive data.

7.2. Compatibility

7.3. Reliability

- Backup systems should be in place to avoid data loss.
- The system shall be hosted on a cloud-based server to achieve 99.9% uptime to ensure availability during school hours.

- The system shall include error handling mechanisms to recover from common issues without interrupting the user experience.

7.4. Scalability

- The system shall be capable of scaling to support additional users and transactions as the school grows.
- The system's database shall support scaling for larger data volumes, including historical financial data, and data will be saved in a non-relational database.

7.5. Maintainability

- Each function of the system shall be implemented in modules to facilitate easy updates and maintenance.
- comprehensive documentation for both users and developers in ensuring that future changes can be made efficiently will be made possible.

7.6. Usability

- The system shall be accessible on both desktop and mobile devices and ensure responsiveness.
- The GUI shall be designed to ensure usability and equity.

7.7. Capacity

7.8. Other non-functional requirements

7.8.1. Performance requirements

<If there are performance requirements for the product under various circumstances, state them here and explain their rationale, to help the developers understand the intent and make suitable design choices. Specify the timing relationships for real time systems. Make such requirements as specific as possible. You may need to state performance requirements for individual functional requirements or features.>

- Page load time should not exceed 3 seconds
- System should support at least 2,000 concurrent users
- The system should load each page within 2 seconds.

- The system should handle many users without performance degradation.

7.8.2. Safety Requirements

<Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. Define any safeguards or actions that must be taken, as well as actions that must be prevented. Refer to any external policies or regulations that state safety issues that affect the product's design or use. Define any safety certifications that must be satisfied.>

- Automatic logout after 15 minutes of inactivity
- Data encryption for all sensitive information

7.8.3. Security Requirements

<Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements. Refer to any external policies or regulations containing security issues that affect the product. Define any security or privacy certifications that must be satisfied.>

7.8.4. Software Quality Attributes

<Specify any additional quality characteristics for the product that will be important to either the customers or the developers. Some to consider are: adaptability, availability, correctness, flexibility, interoperability, maintainability, portability, reliability, reusability, robustness, testability, and usability. Write these to be specific, quantitative, and verifiable when possible. At the least, clarify the relative preferences for various attributes, such as ease of use over ease of learning.>

- Usability: Intuitive UI with a user satisfaction rating of at least 4/5
- Reliability: 99.9% uptime during school hours
- Maintainability: Modular design for easy updates and bug fixes

7.8.5. Legal and regulatory requirement

- The system shall comply with local, regional, and state regulations concerning financial data management and reporting.
- The system shall adhere to accounting standards relevant to educational institutions.

7.8.6. Audit and login requirements

- The system shall log all financial transactions, including what changes were made, and when into a centralized financial digital log book accessible by authorized persons securely.

8. Other requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

< Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Figure: Use case diagram for the student

Figure: Class diagram for the student

Appendix C: Issues List

< This is a dynamic list of the open requirements issues that remain to be resolved, including TBDs, pending decisions, information that is needed, conflicts awaiting resolution, and the like.>